

Experiment : Study of LAN Topology, Network Devices address configuration.

- Objective :
- ① To study LAN topology used in the organization (i.e. college).
 - ② To identify networking devices and their functions.
 - ③ To analyse wired and wireless connections.
 - ④ To measure network speed.
 - ⑤ To find MAC address, IP Address and Subnet Mask.

Theory :- LAN (Local area Network) is a network that connects computer & devices within a limited area such as a laboratory, labs or college campus.

- LAN provides communication and resources sharing between devices. The college mainly uses star Topology because all devices are connected through a central switch.

1. Study of LAN Topology

The college campus network uses a Hybrid Hierarchical Topology. This topology combines:

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- Tree Topology
- Extended star Topology
- Ring Topology (used in Server room)

• The complete network is divided into three hierarchical layers:

- ① core layer
- ② Distribution layer
- ③ Access layer

① core layer

- It is the central part of the Network and is located inside the Server room.

- The Server room contains:

- Firewall
- Router
- Layer-3 Switches
- Patch Rack
- Power Backup/Inverter
- Servers

- This layer controls communication between different parts of the college network and provides internet connectivity.

- Its functions is traffic routing, connects all floors.

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② Distribution Layer

- This layer contains 4-switches.
- The four switches are
 - Main Building (R7) (Room no. 213, 2nd floor)
 - Main Building (R1) (Room no. 301, 3rd floor)
 - R6 (Room no. 213, 1st floor), R2 (Room no. 201, 2nd floor).
- This 4-switches contains :-
 - A port gigabit Easy Smart Switch of 18 port with 16 Port.
 - The 24 Port Gigabit 24 Port 10G/SEF, 12+ managed switches with 24 Port PoE+.

③ Access Layer

- It contains three switches i.e.
 - ① R1 (Room no - 301, 3rd floor)
 - ② R7 (Room no - 05, Ground floor)
 - ③ R5 (Room no - 18, ground floor)
- Every switches act as access layer to which

helps to provide network connectivity to all classroom, library, Academic office, faculty room, lab & etc.

• It connects directly :-

- ① Lab system
- ② Ethernet port.
- ③ wireless AP.
- ④ Outdoor wireless AP (e.g. 218)

• Provides network access to users.

• wired and wireless connectivity.

Server Room Devices

File server	stores files / data
DHCP server	Assign IP addresses
NVR	CCTV Recording
Firewall	network security
Core switch	Backbone connectivity

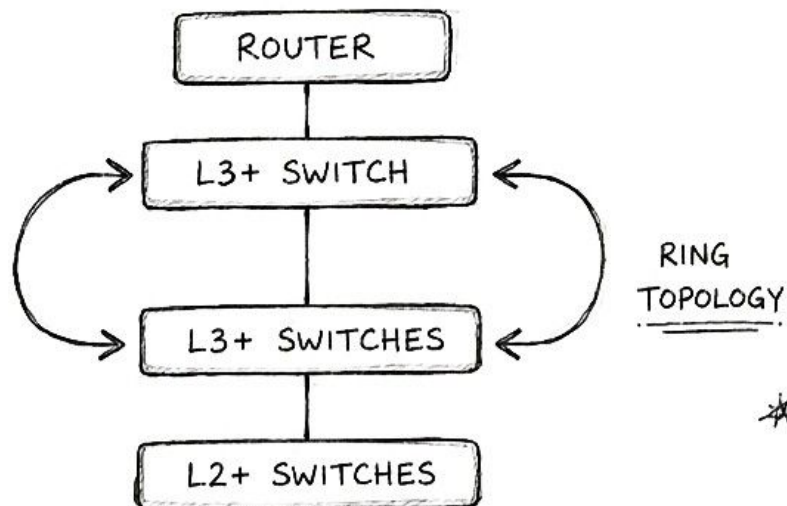
• components used in the topology :-

Router :- connects college network to the Internet

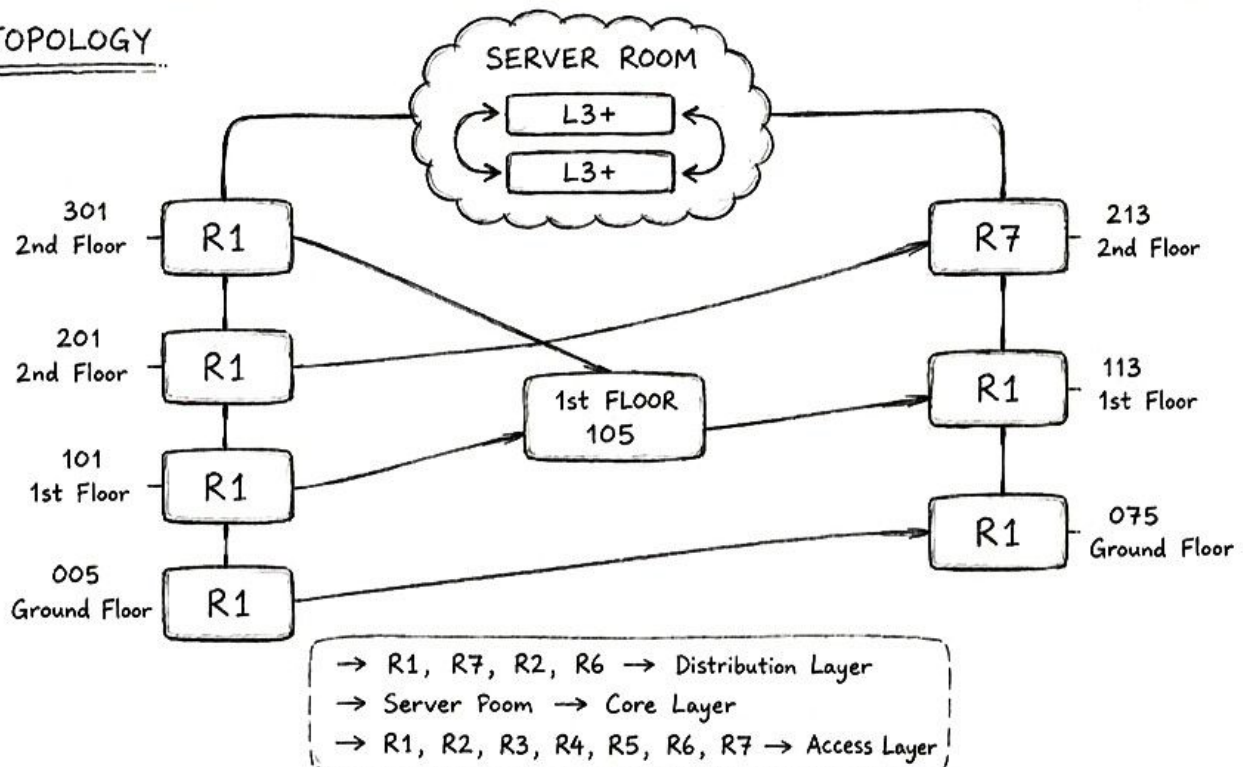
Firewall :- Provides network security & blocks unauthorized access

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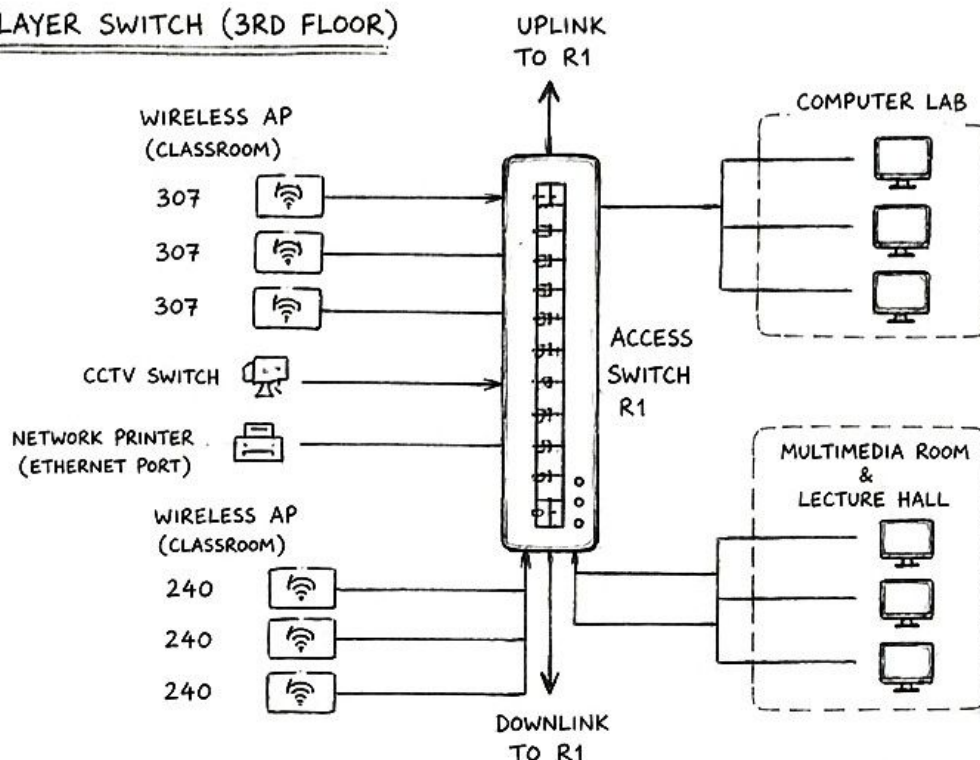
SERVER ROOM ARCHITECTURE / TOPOLOGY



TOPOLOGY



ACCESS LAYER SWITCH (3RD FLOOR)



Performance

Run new task

CPU

5% 2.10 GHz

Memory

4.8/15.7 GB (31%)

Disk 0 (C: D:)

SSD (NVMe)

7%

Wi-Fi

Wi-Fi

S: 88.0 R: 24.0 Kbps

GPU 0

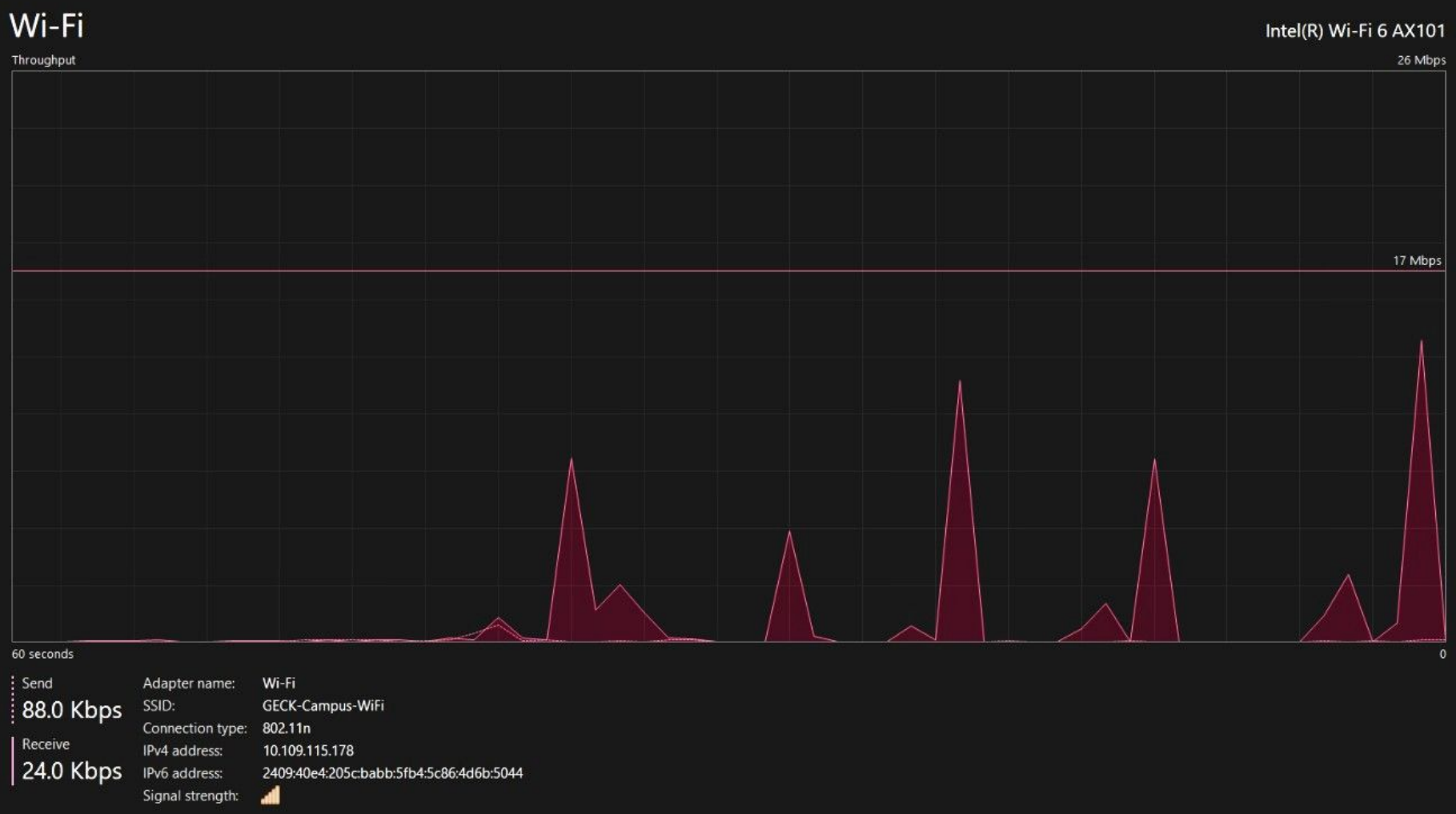
Intel(R) Graphics

1%

GPU 1

NVIDIA GeForce RTX...

0% (40 °C)



```
C:\Windows\system32\cmd.exe: X + v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Krish>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40e4:205c:babb:5fb4:5c86:4d6b:5044
    Temporary IPv6 Address. . . . . : 2409:40e4:205c:babb:89a1:9b8:e6b3:6b3c
    Link-local IPv6 Address . . . . . : fe80::352d:54e5:de82:547c%18
    IPv4 Address. . . . . : 10.109.115.178
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::3073:39ff:fed8:c463%18
                               10.109.115.153

C:\Users\Krish>
```



```
C:\Users\Krish>getmac
```

Physical Address	Transport Name
40-D1-33-04-15-55	\Device\Tcpip_{F5B907E8-1EF0-4B8B-953B-303450EF491E}
D4-93-90-64-B5-D5	Media disconnected

```
C:\Users\Krish>
```


- Core switch : central Device that connects all network segments
- Lab switch : connects computers in computer labs
- Access point (AP) : Provides wireless wifi connection
- Digital Board : smart classroom display connected to network.

3. Analyse between wired & wireless speed & Network connection.

wired network

- It provides stable communication
- Uses Ethernet cable
- Provides high speed & stable connection
- Better security & faster communication

wireless network

- It depends on signal strength.
- Uses wifi signals
- Provides mobility
- It is easy to access & portable.

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• Role of Subnet Mask

- It is used to Identify the network part and host part of an IP address. It helps devices determine whether another device belongs to the same network or a different network.

- Example :-

IP address : 192.000.1.001

Subnet mask : 255.255.000.0

- It is important for:-

- (1) communication b/w devices
- (2) Efficient Routing
- (3) Network Segment

Result

- The LAN topology and networking devices used in the college network were studied successfully.
- wired & wireless communication methods were analyzed and network speed was measured.
- The system network details such as MAC, Address, IP Address and subnet mask were obtained using network commands.

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